

TWO-STAGE GROWTH MODEL

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NET PP&E, DEPRECIATION, CAPITAL EXPENDITURE

Three lines

$$\text{Net PP\&E}_t = k \times \text{Sales}_{t+1}$$

$$\text{Depreciation}_t = d \times \text{Net PP\&E}_{t-1}$$

$$\text{CapEx}_t = \text{Net PP\&E}_t - \text{Net PP\&E}_{t-1} + \text{Depreciation}_t$$

Net PP&E and the depreciation rate are assumptions. Capital expenditure is computed from them.

Capital expenditure as the residual

The asset base is the assumption

Capacity is in place before the sales it serves, so the balance at the end of year t carries year $t + 1$'s sales.

Depreciation follows the balance

A percentage of the opening net balance, not a percentage of sales.

Capital expenditure is the residual

It funds the increase in the balance and replaces what was depreciated during the year.

The zero-growth check

If the balance never changes, capital expenditure equals depreciation.

A FIVE-YEAR PROJECT

Assumptions

ITEM	ASSUMPTION
Sales, year 1	32.00
Sales growth	8% a year
EBITDA margin	25%
Net PP&E	50% of next year's sales
Depreciation	10% of opening net PP&E
Net working capital	10% of next year's sales
Tax rate	21%
Discount rate	12%
Life	5 years, assets sold at book value

The capital schedule

\$ MILLIONS	0	1	2	3	4	5
Sales	—	32.00	34.56	37.32	40.31	43.54
Net PP&E	16.00	17.28	18.66	20.16	21.77	19.59
Depreciation	—	1.60	1.73	1.87	2.02	2.18
Capital expenditure	16.00	2.88	3.11	3.36	3.63	—
Net working capital	3.20	3.46	3.73	4.03	4.35	—

Year 5 buys nothing: there is no year 6 to build for. The closing 19.59 is the year-4 balance less year-5 depreciation. Figures are rounded to 0.01.

The income statement

\$ MILLIONS	1	2	3	4	5
Sales	32.00	34.56	37.32	40.31	43.54
EBITDA	8.00	8.64	9.33	10.08	10.88
Depreciation	1.60	1.73	1.87	2.02	2.18
EBIT	6.40	6.91	7.46	8.06	8.71
Taxes at 21%	1.34	1.45	1.57	1.69	1.83
Net operating profit after tax	5.06	5.46	5.90	6.37	6.88

Add depreciation back

\$ MILLIONS	1	2	3	4	5
Net operating profit after tax	5.06	5.46	5.90	6.37	6.88
Depreciation	1.60	1.73	1.87	2.02	2.18
Operating cash flow	6.66	7.19	7.76	8.38	9.06

Depreciation lowered the tax bill without taking any cash, so it goes back in.

Free cash flow

\$ MILLIONS	0	1	2	3	4	5
Operating cash flow	—	6.66	7.19	7.76	8.38	9.06
Capital expenditure	(16.00)	(2.88)	(3.11)	(3.36)	(3.63)	—
Change in working capital	(3.20)	(0.26)	(0.28)	(0.30)	(0.32)	4.35
Sale at book value, untaxed	—	—	—	—	—	19.59
Free cash flow	(19.20)	3.52	3.80	4.11	4.43	33.00

NPV and IRR

\$11.44m

NPV AT 12%

26.3%

IRR

\$19.20m

INVESTED AT DATE 0

With constant ratios and a sale at book, the IRR is 26.33%, and so is year-1 operating profit after tax over the 19.20 of capital.

THE PROJECT THAT DOES NOT END

Drop the sale, keep the ratios

No sale

The 19.59 of book value stays in the business.

Year 5 keeps investing

Capital expenditure of 3.92 and another 0.35 of working capital, building for year 6.

\$ MILLIONS	1	2	3	4	5
Free cash flow	3.52	3.80	4.11	4.43	4.79
Share of sales	11.0%	11.0%	11.0%	11.0%	11.0%

Every line is a multiple of sales

Constant growth and constant ratios mean free cash flow grows at the sales rate forever.

$$V_0 = \frac{FCF_1}{r - g} = \frac{3.52}{0.12 - 0.08} = 88.00$$

Against 19.20 of capital at date 0, an NPV of 68.80 — against 11.44 when the same business was sold at book in year 5.

THE FIRM

Nothing is bought at date 0

The 16.00 of net PP&E and 3.20 of working capital are already there. No outflow at date 0, so the 88.00 is the value of the firm.

\$88.00m

VALUE OF THE FIRM

11.0x

EBITDA

4.6x

INVESTED CAPITAL

TWO-STAGE GROWTH

The first three years, and then 8%

The same company, with sales that reach 8% only in year 4: 25% growth in year 2, 15% in year 3, and a margin that climbs to 25% over the same three years. It holds 6.00 of cash and 18.00 of debt, against 4.0 million shares.

Years 1 to 3

Sales grow 25% and then 15%. The EBITDA margin climbs from 18% to 21% to 23%.

Year 4 onward

Sales grow 8% a year and the margin holds at 25%. The three ratios are fixed throughout.

Assumptions

ITEM	ASSUMPTION
Sales, year 1	32.00
Sales growth	25%, 15%, then 8% from year 4
EBITDA margin	18%, 21%, 23%, then 25% from year 4
Net PP&E	50% of next year's sales
Depreciation	10% of opening net PP&E
Net working capital	10% of next year's sales
Tax rate	21%
WACC	12%
Cash, debt, shares	6.00, 18.00, 4.0 million

The capital schedule

\$ MILLIONS	0	1	2	3	4	5
Sales	—	32.00	40.00	46.00	49.68	53.65
Next year's sales	32.00	40.00	46.00	49.68	53.65	57.95
Net PP&E	16.00	20.00	23.00	24.84	26.83	28.97
Depreciation	—	1.60	2.00	2.30	2.48	2.68
Capital expenditure	—	5.60	5.00	4.14	4.47	4.83
Net working capital	3.20	4.00	4.60	4.97	5.37	5.79

The date-0 balances are what the company already owns. Figures are rounded to 0.01.

The income statement

\$ MILLIONS	1	2	3	4	5
Sales	32.00	40.00	46.00	49.68	53.65
EBITDA	5.76	8.40	10.58	12.42	13.41
Depreciation	1.60	2.00	2.30	2.48	2.68
EBIT	4.16	6.40	8.28	9.94	10.73
Taxes at 21%	0.87	1.34	1.74	2.09	2.25
Net operating profit after tax	3.29	5.06	6.54	7.85	8.48

Add depreciation back

\$ MILLIONS	1	2	3	4	5
Net operating profit after tax	3.29	5.06	6.54	7.85	8.48
Depreciation	1.60	2.00	2.30	2.48	2.68
Operating cash flow	4.89	7.06	8.84	10.33	11.16

Depreciation lowered the tax bill without taking any cash, so it goes back in.

Free cash flow

\$ MILLIONS	1	2	3	4	5
Operating cash flow	4.89	7.06	8.84	10.33	11.16
Capital expenditure	(5.60)	(5.00)	(4.14)	(4.47)	(4.83)
Increase in working capital	(0.80)	(0.60)	(0.37)	(0.40)	(0.43)
Free cash flow	(1.51)	1.46	4.33	5.46	5.90
Change from prior year	—	n/m	197.6%	26.1%	8.0%
Share of sales	(4.7%)	3.6%	9.4%	11.0%	11.0%

Changes are computed before rounding.

TERMINAL VALUE

Free cash flow in year 5

Free cash flow reaches 11.0% of sales in year 4 and stays there: the margin and all three ratios are fixed from year 4 on. Sales grow 8% in year 5 and every year after, so free cash flow does too.

Year 4 is the first year at the steady ratio. Year 5 is the first year that grows at 8%, which is why the terminal value sits at the end of year 4.

The terminal value

$$TV_4 = \frac{FCF_5}{r - g} = \frac{5.90}{0.12 - 0.08} = 147.55$$

Twenty-five times year-5 free cash flow, at a four-point spread between r and g .

Discount it all back

\$ MILLIONS	1	2	3	4	TV AT 4, YEAR 5 ON
Cash flow	(1.51)	1.46	4.33	5.46	147.55
Discount factor at 12%	0.893	0.797	0.712	0.636	0.636
Present value	(1.35)	1.16	3.08	3.47	93.77

\$100.14m
ENTERPRISE VALUE

94%
IN THE TERMINAL VALUE

Where the 8% comes from

$$g = \text{ROIC} \times \text{reinvestment rate}$$

ROIC

$$8.48 / 32.20 = 26.3\%$$

Reinvestment rate

$$(4.83 + 0.43 - 2.68) / 8.48 = 30.4\%$$

Product

$$26.3\% \times 30.4\% = 8.0\%$$

Year-5 figures, against net PP&E plus working capital of 32.20 at the end of year 4.

CASH AND DEBT

From enterprise value to equity

\$ MILLIONS

Enterprise value	100.14
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Plus cash	6.00
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Less debt	(18.00)
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Equity value	88.14
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Shares outstanding	4.0 million
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Value per share	\$22.03
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Two points about the equity value calculation

Cash is not working capital

The 10% ratio covers receivables, inventory and payables. Cash is added separately, at face value.

Interest is not in free cash flow

The cash flows are unlevered. Debt enters through the WACC and again in the equity value calculation, never in the forecast.

EXERCISE

Value this company

The forecast

Sales of 20.00 in year 1, growing 30%, then 18%, then 6% from year 4 on. EBITDA margin 15%, 18%, 20%, then 22% from year 4 on.

The ratios

Net PP&E 65% of next year's sales.

Depreciation 12% of opening net PP&E.

Working capital 12% of next year's sales.

Tax 21%, WACC 12%.

- 1 Free cash flow for years 1 to 4
- 2 Terminal value at the end of year 4, and the enterprise value
- 3 Equity value per share, with cash of 4.00, debt of 15.00, and 2.5 million shares

JOHN B. SANFILIPPO & SON

The company

Shells, roasts and packages nuts. Fisher, Orchard Valley Harvest, Squirrel Brand, and private label for the grocery chains. Fiscal years end in late June.

Ticker

JBSS, \$805 million of market value: 11.78 million diluted shares at \$68.31.

What it owns

Shelling and roasting plants, \$268 million of net PP&E against \$1.18 billion of sales.

The balance sheet

Simple. The company owns no other businesses, and goodwill is \$11.8 million against \$659 million of assets.

Four years of history

\$ MILLIONS	FY2023	FY2024	FY2025	FY2026
Sales	999.7	1,066.8	1,107.2	1,175.7
Sales growth	—	6.7%	3.8%	6.2%
Operating income	88.9	81.7	82.9	87.0
Operating margin	8.9%	7.7%	7.5%	7.4%
Net PP&E	127.2	178.6	193.0	268.3
Depreciation	22.1	26.6	31.5	33.3
Capital expenditure	20.7	28.3	50.7	88.1

Yahoo returns four years for JBSS, not five, and each ratio needs three.

The three ratios, measured

	FY2023	FY2024	FY2025	FY2026
Net PP&E / next year's sales (k)	11.9%	16.1%	16.4%	—
Depreciation / opening net PP&E (d)	—	20.9%	17.6%	17.3%
Working capital / next year's sales (w)	15.7%	17.1%	21.1%	—

k and w are measured against next year's sales, so they stop at FY2025, the last year with a following year in the data.

Capital expenditure and net PP&E

Three years above depreciation

Capital expenditure of 28.3, 50.7, 88.1 against depreciation of 26.6, 31.5, 33.3. FY2023 was the last year capital expenditure ran below depreciation.

So net PP&E doubles

Net PP&E goes from 127.2 to 268.3 while sales rise 18%. The FY2024 step is a \$59.0m acquisition, almost all of it plant. The FY2026 step is the build.

So k doubles

k goes from 11.9% to 16.4%, and is 21.6% measured against the FY2027 sales forecast.

The forecast has to say whether net PP&E stays this high relative to sales.

The other two ratios

d falls for a mechanical reason

d goes from 20.9% to 17.3%. New assets enter net PP&E faster than the existing assets are depreciated, so *d* falls even though the depreciable lives have not changed. Of the 268.3 balance, 86.9 is construction in progress, which does not depreciate until it is in service. Call *d* 15%.

w moves with inventory

w goes from 15.7% to 21.1%. Inventory alone is 245.8 against net working capital of 220.1, and a nut processor buys a crop once a year. Call *w* 18%.

The forecast

ITEM	ASSUMPTION
Sales growth	5.5%, 5.0%, 4.5%, 3.5%, then 3.0%
Operating margin	7.5%, held
Net PP&E (<i>k</i>)	21.0% of next year's sales
Depreciation (<i>d</i>)	15.0% of opening net PP&E
Working capital (<i>w</i>)	18.0% of next year's sales
Tax rate	25%
WACC	9.0%

k holds near today's 21.6% rather than the FY2025 level of 16.4%. The other rows are read off the history.

The capital schedule

\$ MILLIONS	FY26	FY27	FY28	FY29	FY30	FY31
Sales	1,175.7	1,240.4	1,302.4	1,361.0	1,408.6	1,450.9
Next year's sales	1,240.4	1,302.4	1,361.0	1,408.6	1,450.9	1,494.4
Net PP&E	268.3	273.5	285.8	295.8	304.7	313.8
Depreciation	—	40.2	41.0	42.9	44.4	45.7
Capital expenditure	—	45.4	53.3	52.9	53.2	54.8
Working capital	220.1	234.4	245.0	253.6	261.2	269.0

The FY2026 column is the net PP&E and working capital the company already has.

Free cash flow

\$ MILLIONS	FY27	FY28	FY29	FY30	FY31
EBIT	93.0	97.7	102.1	105.6	108.8
Net operating profit after tax	69.8	73.3	76.6	79.2	81.6
Depreciation	40.2	41.0	42.9	44.4	45.7
Capital expenditure	(45.4)	(53.3)	(52.9)	(53.2)	(54.8)
Increase in working capital	(14.3)	(10.5)	(8.6)	(7.6)	(7.8)
Free cash flow	50.2	50.4	58.0	62.8	64.6

Working capital is positive, so sales growth consumes cash rather than releasing it.

Terminal value

FY2031 grows at 3.0% and all three ratios are fixed, so it is the steady-state year.

$$TV_{2031} = \frac{64.6 \times 1.03}{0.09 - 0.03} = 1,109.6$$

Seventeen times FY2031 free cash flow, at a six-point spread between r and g . The example earlier was twenty-five times, at a four-point spread.

Discount it all back

\$ MILLIONS	FY27	FY28	FY29	FY30	FY31	TV
Free cash flow	50.2	50.4	58.0	62.8	64.6	1,109.6
Discount factor at 9%	0.917	0.842	0.772	0.708	0.650	0.650
Present value	46.1	42.4	44.8	44.5	42.0	721.2

\$940.9m
ENTERPRISE VALUE

77%
IN THE TERMINAL VALUE

From enterprise value to equity

\$ MILLIONS

Enterprise value	940.9
Less borrowings	(83.5)
Plus cash	1.1
Plus goodwill and intangibles	15.2
Less other long-term liabilities	(6.6)
Equity value	867.1
Diluted shares	11.78 million
Value per share	\$73.61

JBSS trades at \$68.31.

What the market is assuming

Hold the five-year forecast fixed and solve for the terminal growth rate that justifies \$68.31.

Implied growth

2.5% forever, against a 3.0% assumption.

Implied exit multiple

6.6 times FY2031 EBITDA.

Today's multiple

7.3 times FY2026 EBITDA.

The price requires the EBITDA multiple to fall from 7.3 times FY2026 to 6.6 times FY2031.

ACCOUNTING

Leases: finance or operating

Does the lease hand over substantially all the risks and rewards of owning the asset?
If so it is a purchase the lessor financed. Any one of five tests settles it.

- 1 Ownership transfers at the end of the term
- 2 There is a purchase option the lessee is reasonably certain to take
- 3 The term covers a major part of the asset's remaining economic life
- 4 The payments are worth substantially all of the asset's fair value
- 5 The asset is so specialised it has no other use to the lessor

Meet none of the five and it is an operating lease.

Leases: what the choice is

Both sit on the balance sheet as a right-of-use asset and a lease liability. A finance lease splits into depreciation and interest; an operating lease is rent in EBIT.

Leave the rent in EBIT

The right-of-use asset stays out of invested capital and the lease liability out of the equity value calculation. The rent is already subtracted in EBIT.

Or capitalise the lease

Take the rent out of EBIT and depreciate the right-of-use asset in its place. The asset joins invested capital and the lease liability joins the equity value calculation.

Either is defensible. Half of the second — the lease liability subtracted in the equity value calculation, the rent still in EBIT — charges twice.

Leases: read the debt line twice

Yahoo reports total debt of 104.1. That is not what the company borrowed.

\$ MILLIONS	
Long-term debt	43.6
Current debt	39.9
Borrowings	83.5
Operating lease liability, non-current	20.6
Yahoo's "total debt"	104.1

No finance leases. The rent is still inside the 87.0 of operating income, so the 20.6 is not subtracted when computing equity value. Subtracting it anyway lowers the value by \$1.75 a share.

Deferred taxes: check, then ignore

	FY2024	FY2025	FY2026
Book effective rate	24.6%	24.3%	25.4%
Cash taxes paid / pretax income	27.7%	27.2%	16.7%

Cash taxes ran 23.8% over the three years against a book rate of 24.8%. Taxing EBIT at the cash rate instead would add about \$1.64 a share.

A one-point gap is small enough to leave the forecast taxing EBIT at the book rate.

Goodwill, intangibles, and other liabilities

Goodwill, \$11.8m

1.8% of assets. No acquisitions are forecast, so goodwill is held flat and added in the equity value calculation, where it adds a dollar a share.

Other intangibles, \$3.4m

Held flat and added in the equity value calculation too, where they add 29 cents a share.

Other long-term liabilities, \$6.6m

The whole “other” category on the balance sheet, and the only one of these three items subtracted in the equity value calculation rather than added.

The three items together move the value per share by 73 cents, against a \$5.30 gap to the market price.

One rule

Every balance sheet item goes in exactly one of two places.

Invested capital

An item goes here if it produces the operating cash flows being discounted. The change in the item is a cash flow in the forecast.

The equity value calculation

Every other item goes here, carried at book value between enterprise value and equity value.

Putting an item in both double counts it; leaving it out of both drops it from the valuation.

Sensitivity to k

	VALUE PER SHARE
k back to 16.4%, the recent plant spending was a one-off	\$80.73
k at 21.0%, the base case	\$73.61
k at 21.6%, today's level holds	\$72.68

d does not appear in this table, and cannot. Capital expenditure less depreciation is the change in the net PP&E balance, and EBIT here is a margin on sales rather than EBITDA less depreciation. So d cancels out of the valuation.

IF THERE IS TIME

Smith-Midland

Precast concrete: highway barriers, sound walls, building panels. **SMID**, \$130 million of market value at \$24.54 a share.

\$ MILLIONS	FY2022	FY2023	FY2024	FY2025
Sales	50.1	59.6	78.5	93.4
Sales growth	—	18.8%	31.8%	19.0%
Operating margin	1.7%	1.9%	12.6%	18.2%
Net PP&E	25.1	27.7	31.7	38.5
Depreciation	2.9	2.4	2.7	2.9
Capital expenditure	2.7	5.0	6.2	9.3

The three ratios, measured

	FY2022	FY2023	FY2024	FY2025
<i>k</i>	42.2%	35.3%	33.9%	—
<i>d</i>	—	9.5%	9.6%	9.1%
<i>w</i>	19.6%	11.2%	14.9%	—

k in the mid-thirties is nearly twice JBSS’s *k*: a concrete plant carries just under three dollars of sales per dollar of net PP&E. *d* holds between 9.1% and 9.6%. The operating margin is the line that moves, from 1.7% to 18.2%.

The forecast

ITEM	ASSUMPTION
Sales growth	12%, 9%, 6%, then 3%
Operating margin	13.0%, down from 18.2%
Net PP&E (<i>k</i>)	36.0% of next year's sales
Depreciation (<i>d</i>)	9.5% of opening net PP&E
Working capital (<i>w</i>)	18.0% of next year's sales
Tax rate, WACC	25%, 11%

Everything but the operating margin is read off the history.

The schedule and the cash flow

\$ MILLIONS	FY25	FY26	FY27	FY28	FY29
Sales	93.4	104.6	114.0	120.9	124.5
Net PP&E	38.5	41.0	43.5	44.8	46.2
Depreciation	—	3.7	3.9	4.1	4.3
Capital expenditure	—	6.2	6.4	5.4	5.6
Working capital	23.2	20.5	21.8	22.4	23.1
Free cash flow	—	10.3	7.4	9.8	10.1

FY2026 working capital falls because the assumed w of 18% is below the 22.2% the company carried at FY2025. A ratio assumption applied to an actual balance always steps in year one.

The value

$$TV_{2029} = \frac{10.1 \times 1.03}{0.11 - 0.03} = 130.3$$

\$ MILLIONS	
Present value of FY26 to FY29	29.2
Present value of terminal value	85.8
Enterprise value	115.0
Plus cash, less debt	11.9, (4.5)
Equity value	122.4
Value per share, on 5.31m shares	\$23.05

The margin implied by the price

\$23.05

AT A 13.0% MARGIN

\$24.54

THE MARKET PRICE

13.7%

THE MARGIN THAT CLOSES THE
GAP

Three quarters of the value is in the terminal value, which rests on the assumed operating margin. The company earned 1.7% in FY2022 and 18.2% in FY2025.

ASSIGNMENT

Flexsteel

Value Flexsteel the way we valued JBSS. Yahoo Finance, ticker **FLXS**, fiscal years ending June 30.

What it does

Makes and sells upholstered furniture. Four years of data, FY2023 through FY2026.

Why this one

All three ratios move over the four years, so each one has to be defended rather than read off the history.

What to pull

STATEMENT	LINES
Income statement	Revenue, operating income, pretax income, tax provision, diluted shares
Balance sheet	Cash, current assets, current liabilities, current debt, net PP&E, total debt, operating lease liability
Cash flow	Depreciation and amortisation, capital expenditure

Net working capital is current assets less cash, minus current liabilities less current debt.

Part 1: the three ratios

Compute, for every year the data allows:

- 1 k , net PP&E as a share of next year's sales
- 2 d , depreciation as a share of opening net PP&E
- 3 w , net working capital as a share of next year's sales

Report each ratio year by year. Two of the three fall by about a third over the four years and the third rises. Say what single fact about the company explains all three, and whether that change is finished.

Part 2: the forecast

ITEM	ASSUMPTION
Sales growth	5.0%, 4.5%, 4.0%, then 3.0%
Operating margin	8.5%, held
Net PP&E (k)	16.0% of next year's sales
Depreciation (d)	5.0% of opening net PP&E
Working capital (w)	16.0% of next year's sales
Tax rate	25%
WACC	10.5%

Use these, not your own averages, so the answers are comparable.

Part 3: the value

- 1 The capital schedule and free cash flow for FY2027 to FY2030
- 2 Terminal value at the end of FY2030, and the enterprise value
- 3 Equity value per share, and the comparison to the market price

FOR THE EQUITY VALUE CALCULATION, AT FY2026		\$ MILLIONS
Borrowings		0.0
Operating lease liability		51.9
Cash		16.7
Share price		\$82.11

The share count is not in the table. Part 4 asks for it.

Part 4: the accounting

Answer in a sentence each.

The depreciable life

Your d will come out near 4%, an implied life above twenty years for a furniture maker's plant. Explain the long implied life, and name the balance sheet line that is the evidence.

The debt line

Yahoo reports total debt of 51.9 and the company has no borrowings. Say what the 51.9 is, whether it belongs in your equity value calculation, and what you would have to change in the forecast to put it there.

Diluted average shares for FY2026 are 5.46 million. Shares outstanding on the cover of the 10-K are 4.11 million. Say which one belongs in the equity value calculation, why, and what happened between the two dates.

What to hand in

The spreadsheet

One tab of history, one of the forecast. Assumptions in their own cells, not buried in formulas.

The ratios

Year by year, with the missing years marked and explained.

One page

Your value per share, the gap to the market price, and the three accounting answers from Part 4.